

Differences in Surveyed Empathy Levels in Scientific Communication after Intervention Training

Client: CSUS Biological Sciences - Shelby Chandar

Course: STAT 192 Statistics Capstone Project

Authors: Austin Melendez, Sara Bruggman

Date: 05/19/2026

1. Summary.....	2
2. Introduction.....	3
2.1 Background.....	3
2.2 Jefferson Scale of Empathy.....	3
2.3 Toronto Empathy Questionnaire.....	4
2.4 Modified Toronto Empathy Questionnaire.....	4
2.5 Dataset.....	4
2.6 Data Preprocessing.....	5
2.6.1 Participant Exclusion.....	5
2.6.2 Standardize and Classify STEM Majors.....	5
2.6.3 Converting Responses to Numeric Values.....	6
2.6.4 Survey Score Reversals.....	6
2.6.5 Calculating Total Survey Score.....	7
2.6.6 Calculate Change Score.....	8
2.6.7 Cleaned Data Outputs.....	8
2.7 Aims of the Report.....	8
3. Preliminary Analysis.....	9
3.1 Data and Group Structure.....	9
3.2 Numerical Summaries.....	10
3.3 Exploratory Visualizations.....	11
3.4 Verification of Assumptions.....	13
3.4.1 Pairing and Independent Observations.....	13
3.4.2 Continuity of Data.....	13
3.4.3 Data Normality.....	13
3.4.4 Heteroscedasticity.....	15
3.4.5 Outliers.....	16
3.4.6 Ceiling-Effect Diagnostics.....	17
4. Methodology.....	19
4.1 Welch's Two-Sample t-Test.....	19
4.2 Paired t-Test.....	20
4.3 Nonparametric Sensitivity Checks.....	21
4.4 Linear Change-Score Models.....	22
4.5 Factorial Change-Score Models.....	22
4.7 ANCOVA.....	23
5. Full-Analysis.....	24
5.1 Baseline STEM vs Non-STEM Differences.....	24
5.2 Overall Pre/Post Change.....	24
5.3 Intervention vs Control Improvement.....	25
5.4 STEM vs Non-STEM Improvement.....	26
5.5 Condition x STEM Interaction.....	27
5.6 Baseline-Adjusted Results.....	29
6. Conclusions.....	31
6.1 Intervention-Specific Conclusion.....	31

6.2 STEM-Specific Conclusion.....	31
6.3 Condition x STEM Interaction Conclusion.....	31
6.4 Results Overview.....	32
6.5 Measurement and Design Limitations.....	32
6.6 Future Recommendations.....	33
7. References.....	33
8. Appendix.....	34
A. Index.....	34
B. Data Dictionary.....	36
C. Alluvial Plots.....	38
D. Tutorial Notebooks.....	39

1. Summary

We examine whether a structured empathy-focused science communication intervention was associated with changes in self-reported empathy among students. Using matched pre- and post-survey data from 222 participants, the analysis evaluates outcomes separately for the Jefferson Scale of Empathy and the Toronto Empathy Questionnaire because the two instruments use different scoring ranges and measure related but distinct aspects of empathy. Participants were compared across Intervention versus Control conditions and STEM versus Non-STEM classifications. The study applied preprocessing procedures to convert Likert responses to numeric scores, reverse-code negatively worded items, compute total empathy scores, and construct participant-level change scores. Statistical methods included Welch two-sample t-tests, paired t-tests, nonparametric sensitivity checks, linear change-score models, factorial interaction models, and ANCOVA models adjusted for baseline empathy.

The strongest finding was a statistically significant overall increase in empathy scores from pre- to post-survey on both measures. Jefferson scores increased by an average of 5.96 points, while Toronto scores increased by an average of 1.31 points. However, the improvement was not significantly greater for Intervention participants than for Control participants. STEM and Non-STEM participants also did not differ significantly in baseline empathy or in average improvement. Factorial change-score models, and baseline-adjusted ANCOVA models found no significant Condition $\times$ STEM interaction, indicating that the Intervention-Control difference in improvement did not depend on STEM classification. Baseline scores were strong predictors of post-survey scores, and ceiling-effect diagnostics suggested that participants with higher initial empathy scores had less room to improve. Overall, the results indicate that self-reported empathy

increased over time, but the evidence does not support attributing that increase specifically to the intervention, STEM classification, or their interaction in this dataset.

2. Introduction

2.1 Background

Empathy is an important part of scientific communication because effective communication depends on understanding an audience's perspective, prior knowledge, and needs. This study evaluates whether a structured, empathy-focused science communication training module is associated with changes in self-reported empathy among students. The analysis uses matched pre and post-survey data and compares outcomes across two primary dimensions.

- Intervention versus Control condition
- STEM versus Non-STEM classification

These comparisons allow us to distinguish general changes in empathy over time from changes that may be specifically associated with the training intervention.

Participants completed two validated empathy measures: the Jefferson Scale of Empathy (JSE) and the Toronto Empathy Questionnaire (TEQ). Because these instruments differ in item composition, response scales, and underlying constructs, all analyses are conducted separately for each measure to ensure appropriate interpretation.

2.2 Jefferson Scale of Empathy

The Jefferson Scale of Empathy (JSE) is designed to measure empathy primarily in professional and interpersonal contexts, with emphasis on perspective-taking and compassionate understanding. In this study, the JSE consists of items J1 through J14. Each item is converted to a numeric value from 1 to 7. In this study, JSE-style items were scored from 1 to 7 and aggregated into a total score, with higher values reflecting greater self-reported empathy.

The standard version of the JSE was used in this study without modification. The total Jefferson score is computed after reverse scoring, and the complete-data score range is 14 to 98.

2.3 Toronto Empathy Questionnaire

The Toronto Empathy Questionnaire (TEQ) is a measure of empathy that focuses on emotional responsiveness and affective components of empathic experience. Unlike the JSE, which emphasizes cognitive and professional perspective-taking, the TEQ captures emotional reactions to others' experiences. In this study, the TEQ consists of items T1 through T16, which are scored from 0 to 4. After being aggregated into a total score, higher values reflect greater self-reported empathy.

A modified version of the TEQ was used to better reflect the context of scientific communication. The total Toronto score is computed after reverse scoring, and the complete-data score range is 0 to 64.

2.4 Modified Toronto Empathy Questionnaire

A modified version of the Toronto Empathy Questionnaire was used in this study to make the survey response format consistent with the Jefferson Scale of Empathy. The original Toronto items were reworded or reformatted so that participants could respond using a Jefferson-style agreement scale rather than the original Toronto response structure.

This modification was intended to unify the response matrix across the survey instrument, making the questionnaire easier for participants to complete and simplifying the data-processing pipeline. After collection, the modified Toronto responses were converted to the numeric Toronto scoring scale used in this report.

Because the Toronto questionnaire was modified from its original format, the Toronto results should be interpreted as scores from a study-specific adapted version of the TEQ rather than as direct scores from the standard TEQ instrument. This limitation is addressed again in [Section 6.5](#).

2.5 Dataset

The analysis uses the anonymized survey dataset, *shelby_anon_data.csv*. The dataset contains student responses collected at two survey times: Pre and Post. Participants were drawn from several undergraduate Biological Sciences course sections in which the empathy-focused science communication program was assigned as part of class activities. However, implementation was not identical across sections: in some course sections, professors treated the program as a required assignment, while in others it was offered as an optional extra-credit opportunity.

Participants are identified using DID #, which allows each student's baseline response to be matched with their follow-up response. In the matched analysis dataset, each retained participant has exactly one Pre survey record and one Post survey record linked by DID #. This matched structure allows within-participant changes in empathy scores to be computed and used in paired tests, change-score models, and baseline-adjusted ANCOVA models.

This course-section implementation context is important for interpretation because participation may not have been fully randomized or equally motivated across all students. In particular, optional extra-credit participation may introduce selection effects, since students who chose to participate may differ from those who did not.

2.6 Data Preprocessing

The preprocessing notebook performs the following data construction steps. The pipeline loads the raw CSV file, removes blank or irrelevant columns, applies the original excluded-ID rule, converts categorical Likert responses to numeric scores, applies reverse scoring, creates total scores, standardizes major labels, derives STEM classification, derives condition labels, and creates matched Pre/Post change scores by participant. These steps ensure consistency in scoring, remove invalid observations, and create variables required for downstream statistical analyses.

2.6.1 Participant Exclusion

Participants were excluded based on a predefined list of flagged IDs provided by the client. The client identified these participants as invalid due to uniformity in their answer patterns throughout the survey. These participants were believed to have clicked through the survey without reading, providing the same answer for every question.

The excluded *DID* # values were 40, 43, 81, 102, 107, 125, 126, 144, and 149. These exclusions were applied prior to scoring or analysis so that all downstream results are based on the same retained sample.

2.6.2 Standardize and Classify STEM Majors

Student majors were first standardized using regular expressions to ensure consistency in naming conventions. Since the survey collected major information using a free-response input field, the resulting data contained inconsistencies such as variations in spelling, capitalization, and terminology. Using regular expressions, these inconsistencies were resolved through a standardization process prior to classification into *STEM* and *Non-STEM*.

Our client provided the following STEM classification guidelines:

- Biology and Biological Science majors are considered *STEM*
- All other majors are considered *Non-STEM*

2.6.3 Converting Responses to Numeric Values

The original text-based survey responses were converted into numeric scores before any analysis was performed. For participant i , item j , and survey time s , let X_{ijs} denote the numeric score assigned to the original response category.

For Jefferson items, responses were scored on a 1-7 scale:

$$X_{ijs} \in \{1, 2, 3, 4, 5, 6, 7\} \quad (2.1)$$

where Extremely Disagree = 1, Disagree = 2, Somewhat Disagree = 3, Neutral = 4, Somewhat Agree = 5, Agree = 6, Extremely Agree = 7.

For Toronto items, responses were scored on a 0-4 scale:

$$X_{ijs} \in \{0, 1, 2, 3, 4\} \quad (2.2)$$

where Disagree = 0, Somewhat Disagree = 1, Neutral = 2, Somewhat Agree = 3, Agree = 4.

2.6.4 Survey Score Reversals

Both empathy instruments include negatively worded items that require reverse scoring to ensure that higher values consistently reflect greater empathy.

For an item scored from lower endpoint L to upper endpoint U , with observed numeric response X , the reverse score is

$$X_{rev} = L + U - X \quad (2.3)$$

For the Jefferson scale, $L = 1$ and $U = 7$, so

$$X_{rev} = 8 - X \quad (2.4)$$

For the Toronto scale, $L = 0$ and $U = 4$, so

$$X_{rev} = 4 - X \quad (2.5)$$

The Jefferson items reverse-coded were $J1$, $J3$, $J6$, $J7$, $J8$, $J11$, and $J13$.

The Toronto items reverse-coded were $T2$, $T3$, $T7$, $T10$, $T11$, $T12$, $T14$, and $T15$.

2.6.5 Calculating Total Survey Score

Two primary outcome variables were constructed for analysis.

For participant i , let J_{ia} be the corrected score for Jefferson item a , after reverse scoring when needed. The Jefferson total is:

$$J_{Total,i} = \sum_{a=1}^{14} J_{ia} \quad (2.6)$$

Computed as the sum of items $J1$ – $J14$, with each item scored on a 1–7 scale.

For participant i , let T_{ib} be the corrected score for Jefferson item b , after reverse scoring when needed. The Toronto total is:

$$T_{Total,i} = \sum_{b=1}^{16} T_{ib} \quad (2.7)$$

Computed as the sum of items $T1$ – $T16$, with each item scored on a 0–4 scale.

All reverse-coded items were adjusted prior to summation. Higher total scores indicate greater self-reported empathy for both measures.

2.6.6 Calculate Change Score

Matched Pre/Post observations were constructed using participant identifiers *DID* #. Only participants with both Pre and Post responses were included in paired analyses.

For each participant i , change scores were computed as:

$$J_{Change} = J_{Post,i} - J_{Pre,i} \quad (2.8)$$

$$T_{Change} = T_{Post,i} - T_{Pre,i} \quad (2.9)$$

These change scores serve as the primary outcome variables for analyses comparing improvement across groups.

2.6.7 Cleaned Data Outputs

The preprocessing pipeline produces three primary datasets for analysis:

- **pre_survey_clean.csv:** contains baseline survey responses and derived variables
- **post_survey_clean.csv:** contains follow-up survey responses and derived variables
- **survey_clean.csv:** contains both the cleaned baseline and follow-up survey data
- **matched_change.csv:** contains matched observations and computed change scores

These datasets form the basis for all subsequent statistical analyses.

2.7 Aims of the Report

1. Document the data-processing and scoring decisions used to create the analysis dataset.
 2. Describe the statistical methods used to evaluate baseline differences, pre/post changes, intervention effects, and subgroup effects.
 3. Determine whether observed changes differ by Intervention versus Control condition and by STEM versus Non-STEM classification.
 4. Assess whether interaction effects suggest differential responsiveness to the training intervention across subgroups.
-

3. Preliminary Analysis

Matching is performed by participant identifier, *DID #*. For each participant and each scale, change is defined above in [Section 2.6.6](#).

All analyses are conducted separately for the Jefferson and Toronto outcomes. This separation is required because the scales have different item counts, scoring ranges, and measurement structures. Jefferson total scores and Toronto total scores should not be directly compared as if they were measured on the same scale. Instead, each scale is interpreted within its own range and scoring framework.

3.1 Data and Group Structure

The analysis uses two between-participant grouping variables. *Condition* identifies whether a participant was in the Control or Intervention arm. *STEM* identifies whether a participant was classified as STEM or Non-STEM under the classification rules outlined in [Section 2.6.2](#)

The matched dataset contains 100 STEM participants and 122 Non-STEM participants. It contains 126 Control participants and 96 Intervention participants. The four Condition by STEM cells are not perfectly balanced, but all four cells contain enough participants to support the planned interaction models:

Condition	STEM group	n
Control	STEM	47
Control	Non-STEM	79
Intervention	STEM	53
Intervention	Non-STEM	43

Table 1. Group class count

These counts matter because the primary interaction question asks whether the Intervention-Control difference in improvement depends on STEM classification. That question

cannot be evaluated from the overall Pre/Post change alone, it requires observations in each Condition $\times$ STEM cell.

3.2 Numerical Summaries

We can summarize means and standard deviations by Survey, Condition, and STEM group. These summaries show that average empathy scores increased from Pre to Post in all four Condition by STEM groups for both scales.

Survey	Condition	STEM group	n	Jefferson mean	Jefferson SD	Toronto mean	Toronto SD
Pre	Control	STEM	47	76.06	10.05	47.66	6.36
Pre	Control	Non-STEM	79	74.90	9.26	47.24	5.84
Pre	Intervention	STEM	53	75.60	9.07	48.17	5.94
Pre	Intervention	Non-STEM	43	77.21	8.83	48.93	5.45
Post	Control	STEM	47	82.60	9.97	49.47	6.41
Post	Control	Non-STEM	79	79.84	11.13	48.52	6.08
Post	Intervention	STEM	53	82.42	10.21	49.26	6.62
Post	Intervention	Non-STEM	43	83.37	9.26	50.00	5.42

Table 2. Summary statistics for average participant total score by group

The change-score summaries also show positive average change in every Condition by STEM group:

Condition	STEM group	n	Jefferson mean change	Toronto mean change
Control	STEM	47	6.53	1.81

Control	Non-STEM	79	4.94	1.28
Intervention	STEM	53	6.81	1.09
Intervention	Non-STEM	43	6.16	1.07

Table 3. Summary statistics for average participant change score by group

Looking at the raw numerical summaries, there is evidence of overall improvement over time, especially for Jefferson scores. However, the descriptive means do not show a large or consistent Intervention advantage over Control, and they do not show a large or consistent Condition by STEM interaction.

3.3 Exploratory Visualizations

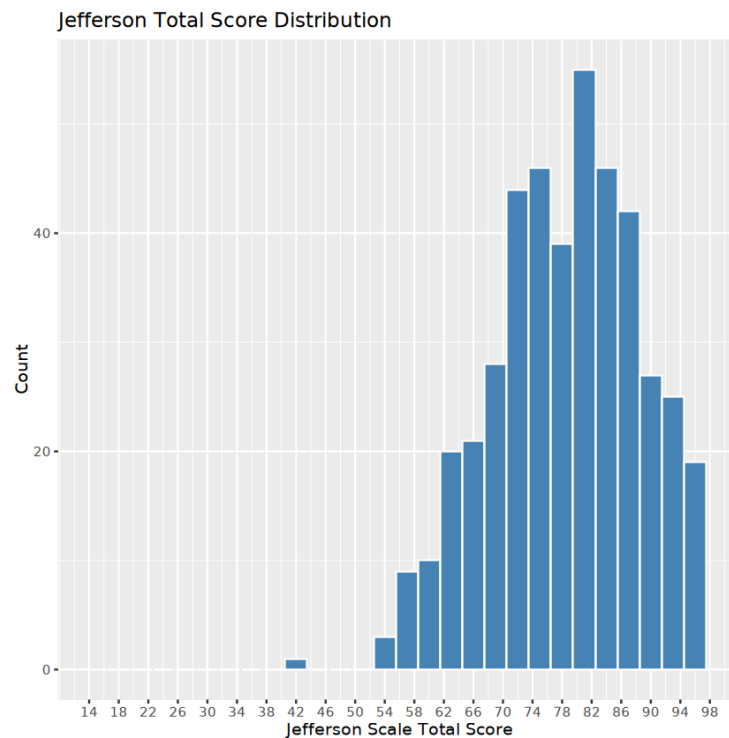


Figure 1. Jefferson scale total score histogram

The Jefferson total score distribution is concentrated toward the upper end of the possible scale range, with most scores falling between approximately 65 and 90. The highest frequencies

appear around the upper 70s to low 80s, suggesting that participants generally reported relatively high empathy on the Jefferson measure. The distribution is slightly left-skewed, with fewer low scores and only a small number of observations below the midrange of the scale.

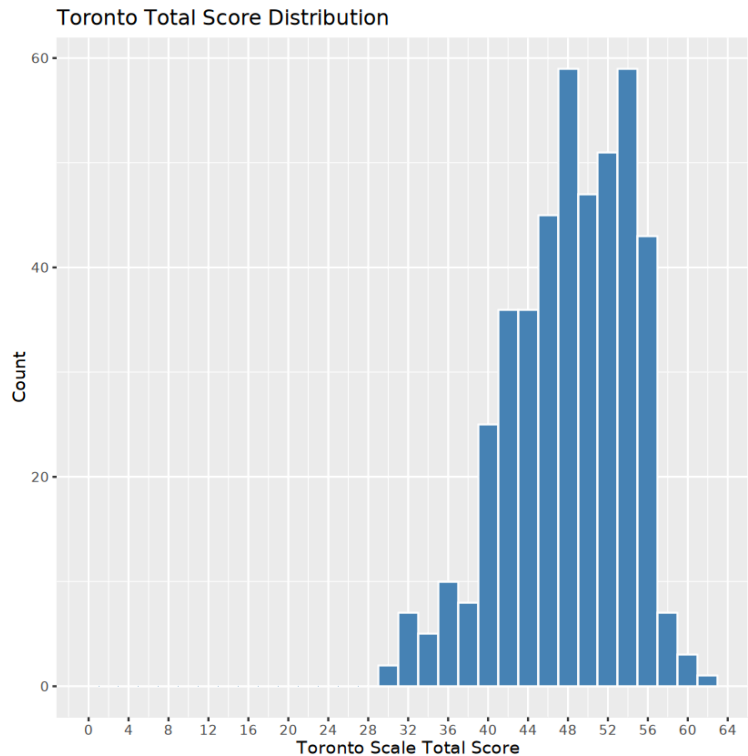


Figure 2. Toronto scale total score histogram

The scores are concentrated mostly between about 40 and 56, with the highest frequency around the upper 40s to mid-50s. The distribution is somewhat left-skewed, meaning there are more scores near the higher end and fewer scores toward the lower end.

We see a pattern indicating a possible ceiling effect, meaning that many participants began or remained near the upper portion of the Jefferson scale, which may limit the amount of observable improvement in later Pre/Post change analyses. We further explore the idea of the ceiling effect in [Section 3.4.6](#).

3.4 Verification of Assumptions

3.4.1 Pairing and Independent Observations

We verify that the retained participants are correctly paired. This check is mathematically necessary because paired t-tests, repeated-measures ANOVA, and change-score models assume that Pre and Post observations are linked within the same participant. If a participant's Pre score were accidentally matched to another participant's Post score, the within-person difference would not represent a real individual change.

For participant i , Y_{0ik} and Y_{1ik} must be measurements from the same participant. The preprocessing notebook creates matched records by *DID #* and verifies alignment before change scores are computed.

The analyses also treat participants as independent statistical units. Formally, the participant-level errors for different students are assumed independent:

$$\varepsilon_i \perp \varepsilon_j \text{ for } i \neq j \quad (3.1)$$

This assumption would be threatened by clustering, such as students nested within teams or sections, if those clusters affected responses. No cluster variable is included in the cleaned modeling dataset, so participant independence is assumed.

3.4.2 Continuity of Data

Summed empathy scores are considered approximately continuous. This is a modeling approximation, not a claim that individual Likert items are continuous. It is more reasonable for a multi-item total than for a single ordinal item because summation creates a wider score range and often produces a distribution closer to continuous measurement.

3.4.3 Data Normality

These Q-Q plots evaluate whether the residuals from the baseline models are approximately normally distributed. We also evaluate whether the residuals from the change-score models are approximately normally distributed.

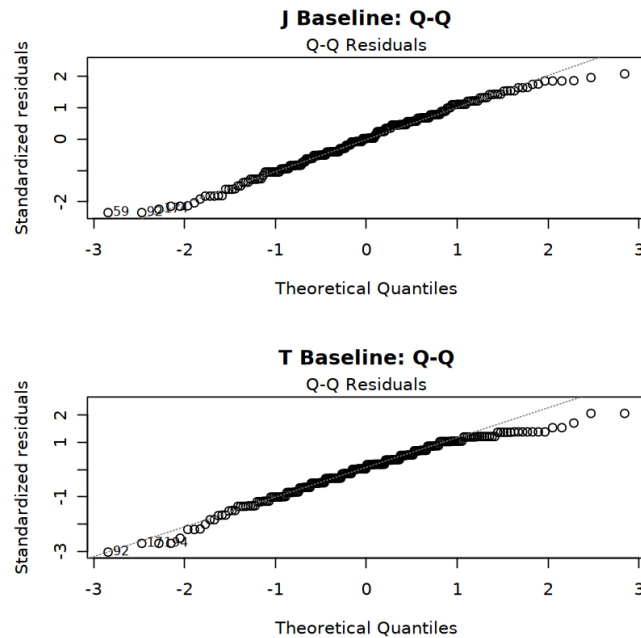


Figure 3. Jefferson and Toronto scale baseline (pre) score Q-Q plot

For the Jefferson baseline model, most points fall close to the reference line, especially in the middle of the distribution. There are mild deviations in the tails, particularly at the lower and upper extremes, but the overall pattern looks reasonably normal. For the Toronto baseline model, the middle of the distribution also follows the line fairly well, with similar mild deviations in the tails. Overall, the baseline residuals appear approximately normal, with some tail deviations. These deviations are not unusual in survey total-score data and do not appear severe enough, by themselves, to invalidate the use of linear models or t-based procedures.

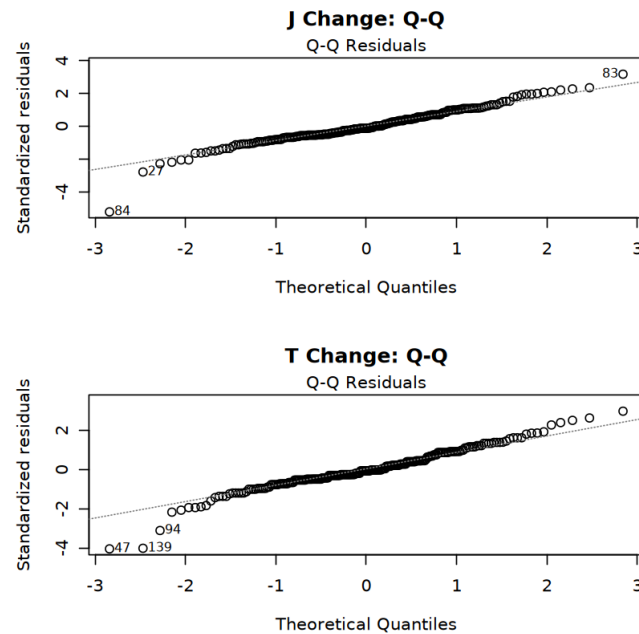


Figure 4. Jefferson and Toronto scale change score Q-Q plot

For Jefferson change, the middle of the distribution follows the reference line reasonably well, but the tails deviate more noticeably. The lower tail includes a large negative residual, while the upper tail includes a positive residual. This suggests that Jefferson change scores have some outliers or heavier tails than a perfect normal distribution.

For Toronto change, the middle of the distribution is fairly linear, but there are several large negative residuals in the lower tail. This also indicates heavier tails or outlying change scores.

3.4.4 Heteroscedasticity

Levene's tests using the four Condition by STEM cells provide additional evidence:

H_0 : The population variances are equal

H_1 : At least one σ_k^2 is different from the others (3.2)

If $p < 0.05$, H_0 is rejected, indicating unequal variances.

Outcome	Levene statistic	p-value

Jefferson change score	0.542	0.654
Toronto change score	0.460	0.711
Jefferson total score	0.651	0.583
Toronto total score	0.538	0.657

Table 4. Levene's tests using the four Condition $\times$ STEM combinations

Therefore the null hypothesis is retained, we can conclude the population variances are approximately equal.

3.4.5 Outliers

For Jefferson change, the most extreme case is DID #87 in the Control/Non-STEM group. This participant decreased from $J_{Pre} = 74$, $J_{Post} = 41$, producing $J_{Change} = -33$. In the factorial Jefferson change model, this same observation has the largest Cook's distance, approximately 0.088. Using the common screening rule $\frac{4}{n}$, the comparison value is $\frac{4}{222} = 0.018$, so DID #87 is influential enough to review for response consistency or data-entry validity. Its standardized residual is also large, approximately -5.23. This does not by itself justify removal, but it should be treated as a sensitivity concern.

Toronto change scores show a smaller average Pre-to-Post shift overall, so large individual changes are more visible relative to the group distributions. The largest negative Toronto change cases are DID #147 in Intervention/STEM and DID #49 in Control/Non-STEM, each with $T_{Change} = -16$. These are also the two largest Cook's-distance cases in the Toronto factorial change model, with Cook's distances of approximately 0.078 and 0.053, respectively.

The current analysis retains these observations because the outlier diagnostics do not establish that the values are erroneous, but further investigation is warranted. There are a small number of large individual shifts, specifically in DID #87 for Jefferson and DID #147 and DID #49 for Toronto, which contribute noticeably to residual and influence diagnostics, but the primary pattern remains that the formal Condition, STEM, and Condition by STEM effects are not statistically significant in the fitted models.

3.4.6 Ceiling-Effect Diagnostics

Because baseline empathy scores are high relative to each scale range, ceiling effects are a serious concern. If participants start near the upper limit of a scale, they have less room to improve, even if the intervention is helpful. The ceiling-effect model is

$$\Delta_{ik} = \eta_{0k} + \eta_{1k} Y_{0ik} + e_{ik} \quad (3.3)$$

A negative η_{1k} indicates that higher baseline scores predict smaller improvements.

Scale	Baseline-change correlation	Baseline slope	p-value
Jefferson	-0.236	-0.186	0.000384
Toronto	-0.302	-0.219	0.00000466

Table 5. Ceiling effect statistics

Both scales show a statistically significant negative relationship between baseline score and change. For Jefferson, the baseline-change correlation is -0.236, the slope is -0.186, and the p-value is 0.000384. For Toronto, the baseline-change correlation is -0.302, the slope is -0.219, and the p-value is 0.00000466.

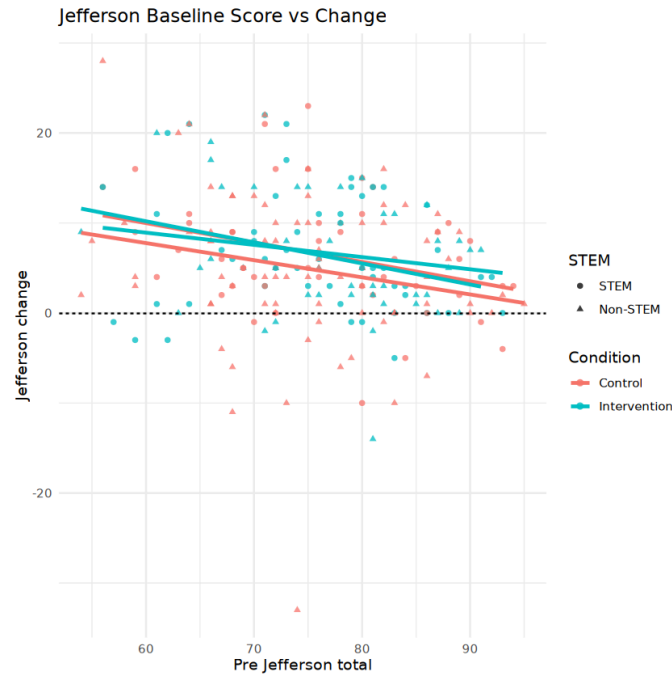


Figure 5. Jefferson baseline (pre) total score vs change score

The main pattern is a negative relationship. Participants with higher baseline Jefferson scores tended to show smaller gains. We can see that both fitted trend lines slope downward.

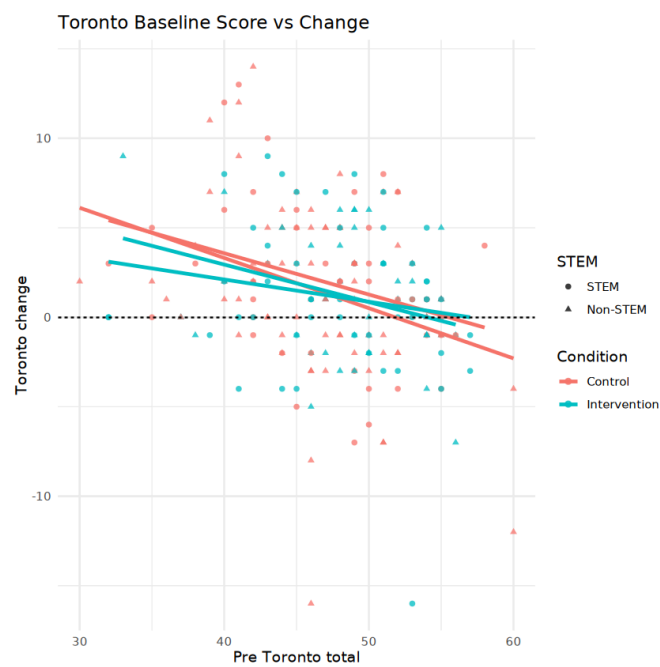


Figure 6. Toronto baseline (pre) total score vs change score

Like the Jefferson plot, the fitted lines slope downward, indicating that participants with higher baseline Toronto scores tended to improve less. The Control line appears slightly steeper than the Intervention line, but the two lines are fairly close.

These results mean that participants with higher baseline empathy tended to improve less. That is consistent with a ceiling-effect pattern. This does not invalidate the paired Pre/Post improvement finding, but it does affect interpretation. Raw change scores only partly reflect how much room participants had to improve. Because of this, the full analysis reports ANCOVA models that estimate Post scores while adjusting for baseline scores.

4. Methodology

The statistical analysis is organized around five research questions:

1. Do STEM and Non-STEM students differ in baseline empathy?
2. Does self-reported empathy change from Pre to Post?
3. Do Intervention participants improve differently from Control participants?
4. Do STEM and Non-STEM students improve differently?
5. Does the Intervention-Control difference in improvement depend on STEM status?

The analysis uses a sequence of related models rather than a single test. This is appropriate because the study contains both within-participant structure, represented by Pre and Post scores, and between-participant structure, represented by Condition and STEM classification. The participant's total baseline score was calculated based on the equations in [Section 2.6.5](#). The participant's change score was calculated based on the equations in [Section 2.6.6](#).

4.1 Welch's Two-Sample t-Test

Welch's two-sample t-test was used to compare the mean response between two independent groups when the population variances cannot be assumed to be equal. This test is appropriate when the goal is to determine whether two groups have different average outcomes.

For a given outcome, the null hypothesis is:

$$H_0: \mu_A - \mu_B = 0$$

or equivalently:

$$H_0: \mu_{\Delta} = 0$$

where μ_A and μ_B are the population means for the two groups being compared. Welch's test was preferred over the pooled-variance t-test because the analysis cells were not perfectly balanced and the assumption checks indicated possible heteroscedasticity. This makes Welch's test the more conservative and appropriate default for independent comparisons.

$$t_W = \frac{\bar{Y}_A - \bar{Y}_B}{\sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}}} \quad (4.1)$$

where $\bar{Y}_A$ and $\bar{Y}_B$ are the sample means, s_A^2 and s_B^2 are the sample variances, and n_A and n_B are the sample sizes for the two groups.

The numerator is the observed difference between the two sample means. The denominator is the estimated standard error of the difference in means. If the p-value is less than the chosen significance level, typically $\alpha = 0.05$, the null hypothesis is rejected. This would provide statistical evidence that the two groups differ in their mean outcome scores. If the p-value is greater than or equal to α , there is insufficient evidence to conclude that the population means are different.

4.2 Paired t-Test

A paired t-test was used to compare the mean outcome measured on the same observational units under two related conditions. This test is appropriate when each observation in one group is naturally matched with an observation in the other group. Because the two measurements come from the same individual, the observations are not independent across time. The paired t-test accounts for this dependence by analyzing the within-subject differences rather than treating the two samples as unrelated groups.

The null hypothesis for the paired t-test is that the population mean difference is zero:

$$H_0: \mu_A - \mu_B = 0$$

or equivalently:

$$H_0: \mu_{\Delta} = 0$$

where μ_A and μ_B are the population means for the two groups being compared.

The paired t-test evaluates whether the mean within-participant change differs from zero. The paired t-statistic is:

$$t_p = \frac{\bar{\Delta}}{s_{\Delta} / \sqrt{n}} \quad (4.2)$$

where $\bar{\Delta}$ is the sample mean change score, s_{Δ} is the standard deviation of the change scores, and n is the number of paired observations.

The p-value is calculated from the t-distribution with $n - 1$ degrees of freedom. If the p-value is less than the chosen significance level, typically $\alpha = 0.05$, the null hypothesis is rejected. This indicates that there is statistical evidence of a nonzero average change between the paired measurements. If the p-value is greater than or equal to α , there is insufficient evidence to conclude that the average difference differs from zero.

4.3 Nonparametric Sensitivity Checks

The notebooks included nonparametric sensitivity checks to assess whether the main conclusions depended on parametric assumptions. Wilcoxon rank-sum tests were used for independent baseline comparisons, and Wilcoxon signed-rank tests were used for paired Pre/Post comparisons.

These checks are appropriate because Jefferson and Toronto scores are summed survey-scale scores derived from item responses. Although summed multi-item scores are commonly analyzed with mean-based models, nonparametric checks help verify that conclusions are not driven by strong distributional assumptions.

The nonparametric tests were treated as sensitivity checks, not replacements for the main models. The primary interpretation remains based on the Welch tests, paired t-tests, linear models, and ANCOVA because those methods directly estimate the mean differences and interactions central to the research questions.

4.4 Linear Change-Score Models

Linear change-score models were used to estimate whether average improvement differed across participant groups. The outcome variable in these models was the participant-level change score defined in [Section 2.6.6](#).

For simple two-group comparisons, the linear change-score model was written as:

$$\Delta_i = \beta_0 + \beta_1 G_i + \varepsilon_i \quad (4.3)$$

where Δ_i is the change score for participant i , G_i is an indicator variable for the group comparison of interest, β_0 is the intercept, β_1 is the group effect and ε_i is the error term.

These models are useful because they express improvement as a single participant-level outcome. They also connect directly to the Welch change-score comparisons while allowing extensions to factorial models.

4.5 Factorial Change-Score Models

The primary interaction question was tested using factorial change-score models. These models were used to determine whether the difference in improvement between the Intervention and Control conditions depended on STEM classification. The outcome variable was the participant-level change score defined in [Section 2.6.6](#).

$$\Delta_i = \beta_0 + \beta_1 C_i + \beta_2 S_i + \beta_3 (C_i S_i) + \varepsilon_i \quad (4.4)$$

where Δ_i is the change score for participant i , C_i is the condition indicator, S_i is the STEM classification indicator, $C_i S_i$ is the interaction between condition and STEM classification, and ε_i is the error term.

β_0 represents the average change score for participants in the Control/STEM reference group. The coefficient β_1 represents the difference in mean change between Intervention and Control participants among STEM participants. This is because when $S_i = 0$, the interaction term $C_i S_i = 0$. The coefficient β_2 represents the Non-STEM minus STEM difference among Control participants. This is because when $C_i = 0$, the interaction term $C_i S_i = 0$. The coefficient β_3 is the interaction coefficient. It measures whether the Intervention-Control difference in average

improvement differs between Non-STEM and STEM participants. Thus, β_3 compares the Intervention-Control difference among Non-STEM participants to the Intervention-Control difference among STEM participants.

4.7 ANCOVA

Analysis of covariance, or ANCOVA, was used to compare post-intervention scores across groups while adjusting for baseline differences in pre-intervention scores. This approach is appropriate when the outcome of interest is the post-intervention score, but participants may have entered the study with different baseline scores. By including the pre-intervention score as a covariate, the model estimates group differences in post scores after accounting for initial score levels.

For the Condition $\times$ STEM analysis, the ANCOVA model was written as:

$$Y_{i1} = \beta_0 + \beta_1 Y_{i0} + \beta_2 C_i + \beta_3 S_i + \beta_4 (C_i S_i) + \varepsilon_i \quad (4.6)$$

where Y_{i1} is the post-intervention score for participant i , Y_{i0} is the pre-intervention score for participant i , C_i is the condition indicator, S_i is the STEM classification indicator, $C_i S_i$ is the Condition $\times$ STEM interaction term, and ε_i is the participant-level error term.

The coefficient β_1 adjusts for baseline differences in pre-intervention scores. The coefficient β_2 represents the adjusted Intervention-Control difference among STEM participants. The coefficient β_3 represents the adjusted Non-STEM minus STEM difference among Control participants. The coefficient β_4 is the adjusted Condition $\times$ STEM interaction. It tests whether the adjusted Intervention-Control difference in post-intervention scores differs between STEM and Non-STEM participants.

The corresponding null hypothesis for the interaction is:

$$H_0: \beta_4 = 0$$

This means that, after adjusting for baseline scores, the Intervention-Control difference in post-intervention scores is the same for STEM and Non-STEM participants.

5. Full-Analysis

5.1 Baseline STEM vs Non-STEM Differences

Model: `t.test(J_total ~ STEM)`

At baseline, STEM and Non-STEM participants had very similar empathy scores.

For Jefferson, STEM participants had a baseline mean of 75.82, while Non-STEM participants had a baseline mean of 75.71. Welch's test found no statistically significant difference:

Comparison	Test	t	df	p	95% CI
STEM vs Non-STEM	Welch two-sample <i>t</i> -test	-0.085	208.23	0.933	[-2.59, 2.38]

Table 6. Welch's two-sample *t*-test Jefferson baseline scores (STEM vs Non-STEM)

Model: `t.test(T_total ~ STEM)`

For Toronto, STEM participants had a baseline mean of 47.93, while Non-STEM participants had a baseline mean of 47.84. Welch's test again found no statistically significant difference:

Comparison	Test	t	df	p	95% CI
STEM vs Non-STEM	Welch two-sample <i>t</i> -test	-0.117	205.78	0.907	[-1.68, 1.49]

Table 7. Welch's two-sample *t*-test Toronto baseline scores (STEM vs Non-STEM)

Thus, we see no evidence of meaningful baseline STEM versus Non-STEM differences on either scale.

5.2 Overall Pre/Post Change

Model: `t.test(pre_survey_clean$J_total, post_survey_clean$J_total, paired = TRUE)`

The paired Pre/Post analysis found statistically significant improvement on both scales. For Jefferson, the mean Post minus Pre change was +5.96 points. The paired *t*-test was significant:

Comparison	Test	t	df	p	95% CI	Cohen's d

Pre vs Post	Paired-samples <i>t</i> -test	-12.17	221	< .001	[4.99, 6.92]	0.817
-------------	-------------------------------	--------	-----	--------	--------------	-------

Table 8. Paired *t*-test Jefferson change scores (Pre vs Post)

The R output reports the test as Pre minus Post, which is why the *t* statistic is negative. Interpreted as Post minus Pre, the result is a positive improvement.

Model: `t.test(pre_survey_clean$T_total, post_survey_clean$T_total, paired = TRUE)`

For Toronto, the mean Post minus Pre change was +1.31 points:

Comparison	Test	t	df	p	95% CI	Cohen's d
Pre vs Post	Paired-samples <i>t</i> -test	-4.55	221	< .001	[0.74, 1.87]	0.306

Table 9. Paired *t*-test Toronto change scores (Pre vs Post)

These results show that self-reported empathy increased from Pre to Post across the matched sample, with a larger paired effect for Jefferson than for Toronto.

5.3 Intervention vs Control Improvement

Model: `t.test(J_change ~ Condition)`

The Intervention group did not improve significantly more than the Control group.

For Jefferson, the Intervention group improved by 6.52 points on average, while the Control group improved by 5.53 points. The difference was about 0.99 points and was not statistically significant:

Comparison	Test	t	df	p	95% CI
Intervention vs Control	Welch two-sample <i>t</i> -test	1.02	217.11	0.308	[-0.92, 2.90]

Table 10. Welch's two-sample *t*-test Jefferson change scores (Intervention vs Control)

Model: `t.test(T_change ~ Condition)`

For Toronto, the Intervention group improved by 1.08 points, while the Control group improved by 1.48 points. The difference was about -0.39 points and was not statistically significant:

Comparison	Test	t	df	p	95% CI
Intervention vs Control	Welch two-sample <i>t</i> -test	-0.69	217.75	0.489	[-1.51, 0.72]

Table 11. Welch's two-sample *t*-test Toronto change scores (Intervention vs Control)

Although scores increased over time overall, we do not see evidence that improvement was specific to the Intervention group. Therefore, we can say the Survey x Condition interaction was not significant for either scale.

5.4 STEM vs Non-STEM Improvement

Model: `t.test(J_change ~ STEM)`

STEM and Non-STEM participants did not differ significantly in improvement.

For Jefferson, STEM participants improved by 6.68 points on average, while Non-STEM participants improved by 5.37 points. The difference was not statistically significant:

Comparison	Test	t	df	p	95% CI
STEM vs Non-STEM	Welch two-sample <i>t</i> -test	1.36	219.92	0.176	[-0.59, 3.21]

Table 12. Welch's two-sample *t*-test Jefferson change scores (STEM vs Non-STEM)

Model: `t.test(T_change ~ STEM)`

For Toronto, STEM participants improved by 1.43 points, while Non-STEM participants improved by 1.20 points. This difference was also not statistically significant:

Comparison	Test	t	df	p	95% CI
STEM vs Non-STEM	Welch two-sample <i>t</i> -test	0.39	212.13	0.697	[-0.91, 1.36]

Table 13. Welch's two-sample t-test Toronto change scores (STEM vs Non-STEM)

Therefore, we can say the Survey x STEM interaction was not significant for either scale.

5.5 Condition x STEM Interaction

The factorial change-score models directly tested whether the Intervention-Control difference in improvement depended on STEM status.

Model: $\text{lm}(J_change \sim \text{Condition} * \text{STEM})$

For Jefferson, the model used Control/STEM as the reference group. The results were:

Predictor	Estimate	p	t	Std. Error
Intercept	6.53	< .001	6.13	1.07
Intervention	0.28	.849	0.19	1.46
Non-STEM	-1.60	.237	-1.19	1.35
Condition × STEM	0.95	.639	0.47	2.01
R^2	0.012		Model explains approximately 1.2% of Jefferson change-score variability.	

Table 14. Jefferson Condition × STEM Interaction Model

A anova: 4 × 5					
	Df	Sum Sq	Mean Sq	F value	Pr(>F)
	<int>	<dbl>	<dbl>	<dbl>	<dbl>
Condition	1	53.30379	53.30379	0.9998164	0.3184635
STEM	1	73.18904	73.18904	1.3728031	0.2426098
Condition:STEM	1	11.78297	11.78297	0.2210125	0.6387395
Residuals	218	11622.35934	53.31357	NA	NA

Figure 7. Jefferson Condition $\times$ STEM ANOVAModel: $\text{lm}(\text{T_change} \sim \text{Condition} * \text{STEM})$

For Toronto, the results were:

Predictor	Estimate	p	t	Std. Error
Intercept	1.81	.004	2.89	0.63
Intervention	-0.71	.407	-0.83	0.86
Non-STEM	-0.53	.504	-0.67	0.79
Condition $\times$ STEM	0.51	.670	0.43	1.18
R^2	0.004		Model explains approximately 0.4% of Toronto change-score variability.	

Table 15. Toronto Condition $\times$ STEM Interaction Model

A anova: 4 $\times$ 5					
	Df	Sum Sq	Mean Sq	F value	Pr(>F)
	<int>	<dbl>	<dbl>	<dbl>	<dbl>
Condition	1	8.409266	8.409266	0.4559717	0.5002282
STEM	1	4.933806	4.933806	0.2675234	0.6055223
Condition:STEM	1	3.359086	3.359086	0.1821381	0.6699633
Residuals	218	4020.469013	18.442518	NA	NA

Figure 8. Toronto Condition $\times$ STEM ANOVATherefore, we can say the Condition $\times$ STEM interaction was not significant for either scale.

5.6 Baseline-Adjusted Results

ANCOVA models showed that baseline score strongly predicted Post score, but Condition, STEM, and their interaction were not significant after baseline adjustment.

Model: $\text{lm}(J_post \sim J_pre + \text{Condition} * \text{STEM})$

For Jefferson:

Predictor	Estimate	p	t	Std. Error
Intercept	21.11	< .001	5.20	4.06
Baseline Pre score	0.808	< .001	15.65	0.052
Intervention	0.19	.893	0.13	1.42
Non-STEM	-1.82	.166	-1.39	1.31
Condition $\times$ STEM	1.48	.452	0.75	1.96
R^2	0.539		Model explains approximately 53.9% of Jefferson post-score variability	

Table 16. Jefferson Baseline-Adjusted Model

A anova: 5 $\times$ 5					
	Df	Sum Sq	Mean Sq	F value	Pr(>F)
	<int>	<dbl>	<dbl>	<dbl>	<dbl>
J_pre	1	12620.61403	12620.61403	250.5917008	4.923717e-38
Condition	1	75.16457	75.16457	1.4924485	2.231624e-01
STEM	1	71.52281	71.52281	1.4201388	2.346822e-01
Condition:STEM	1	28.55668	28.55668	0.5670143	4.522639e-01
Residuals	217	10928.82659	50.36326	NA	NA

Figure 9. Jefferson baseline adjusted ANCOVA

Model: $\text{lm}(T_{\text{post}} \sim T_{\text{pre}} + \text{Condition} * \text{STEM})$

For Toronto:

Predictor	Estimate	p	t	Std. Error
Intercept	12.24	< .001	5.27	2.32
Baseline Pre score	0.781	< .001	16.59	0.047
Intervention	-0.60	.465	-0.73	0.82
Non-STEM	-0.62	.412	-0.82	0.76
Condition $\times$ STEM	0.76	.501	0.67	1.13
R^2	0.563		Model explains approximately 56.3% of Toronto post-score variability	

Table 17. Toronto Baseline-Adjusted Model

A anova: 5 $\times$ 5

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
	<int>	<dbl>	<dbl>	<dbl>	<dbl>
T_pre	1	4692.144281	4692.144281	278.48437691	8.952048e-41
Condition	1	1.224528	1.224528	0.07267722	7.877341e-01
STEM	1	4.207428	4.207428	0.24971586	6.177815e-01
Condition:STEM	1	7.649070	7.649070	0.45398144	5.011673e-01
Residuals	217	3656.202622	16.848860	NA	NA

Figure 10. Toronto baseline adjusted ANCOVA

These results indicate that baseline empathy was the strongest predictor of Post empathy. Adjusting for baseline did not reveal an Intervention effect, STEM effect, or Condition $\times$ STEM interaction.

6. Conclusions

6.1 Intervention-Specific Conclusion

The data does not provide strong evidence that the Intervention group improved more than the Control group. As shown in [Section 5.3](#), the simple change-score comparisons did not find statistically significant Intervention-Control differences for either Jefferson or Toronto scores. Improvement over time was not clearly intervention-specific after accounting for the study design and baseline scores.

Therefore, the most defensible conclusion is that empathy scores increased from Pre to Post, but the observed increase cannot be attributed specifically to the Intervention condition.

6.2 STEM-Specific Conclusion

The data also does not provide strong evidence that STEM and Non-STEM participants differed in baseline empathy or in improvement over time. As shown in [Section 5.1](#), baseline Jefferson and Toronto scores were very similar between STEM and Non-STEM participants. As shown in [Section 5.4](#), STEM and Non-STEM participants also did not differ significantly in change scores.

Thus, under the project classification rules outlined in [Section 2.6.2](#), STEM status was not a strong predictor of baseline empathy or Pre-to-Post improvement in this dataset.

6.3 Condition x STEM Interaction Conclusion

The Condition $\times$ STEM interaction was not statistically significant for either Jefferson or Toronto scores. As shown in [Section 5.5](#), the factorial change-score models did not provide evidence that the Intervention-Control difference in improvement depended on STEM status. The baseline-adjusted ANCOVA results reported in [Section 5.6](#) reached the same general conclusion.

This means that the study did not find evidence that STEM and Non-STEM participants responded differently to the Intervention relative to the Control condition.

6.4 Results Overview

The strongest and most consistent finding was an overall improvement from Pre to Post. As reported in [Section 5.2](#), both Jefferson and Toronto scores increased significantly in the matched sample of 222 participants. The Jefferson scale showed a larger within-person improvement than the Toronto scale, but the two measures should not be directly compared as if they were measured on the same unit.

This overall improvement should not be interpreted as evidence of an intervention-specific effect by itself. Control participants also improved, and the models comparing Intervention and Control change did not find statistically significant differences. Across the Welch change-score tests, factorial change-score models, and ANCOVA, the same general pattern appeared: scores increased over time, but the increase was not clearly attributable to Condition, STEM status, or their interaction.

6.5 Measurement and Design Limitations

Several measurement and design limitations affect interpretation. First, Jefferson and Toronto scores must be interpreted separately because they use different scoring ranges and measurement structures. Although the Jefferson improvement was larger than the Toronto improvement, the two scales should not be treated as directly comparable measures.

Second, the outcomes are self-reported empathy scores. Self-report measures can capture meaningful changes in perceived empathy, but they may also reflect response tendencies, social desirability, or increased familiarity with the survey.

Third, the notebooks found evidence of ceiling effects, as discussed in [Section 3.4.6](#). Participants with higher baseline scores tended to show smaller gains, suggesting that some students had limited room to improve on the scales. This is why the ANCOVA models in [Section 5.6](#) are important companion analyses.

Fourth, the modifications to the Toronto Empathy Questionnaire described in [Section 2.4](#) may have introduced additional measurement variability because the original TEQ response structure was adapted to match the study-specific survey format. As a result, Toronto scores in this report should be interpreted as coming from a study-specific modified TEQ rather than the standard validated TEQ instrument.

Finally, the Condition $\times$ STEM cells were usable but not perfectly balanced, as shown in [Section 3.1](#). The imbalance does not invalidate the models, but it limits precision when estimating interaction effects. As a result, non-significant interaction findings should be interpreted as lack

of evidence for an interaction in this dataset, not as proof that no interaction could exist in a larger or more balanced study.

6.6 Future Recommendations

Future studies should use a more controlled experimental design to better isolate the effect of the intervention. Ideally, participants should be randomly assigned to Intervention and Control groups across comparable course sections. If random assignment is not possible, future analyses should collect additional background variables and use adjustment methods to account for differences between groups.

The survey instruments should also be validated for this specific study context. In particular, the modified Toronto scale should be tested for reliability and consistency after changing the response structure. Reporting internal consistency measures, such as Cronbach's alpha, would help determine whether the items function as a coherent empathy scale.

Finally, future studies should increase sample size and improve follow-up completion. Administering Pre and Post surveys under consistent conditions, making participation requirements uniform across sections, and reducing DID entry errors would improve data quality. A larger and more balanced sample would also make it easier to compare STEM and Non-STEM students, evaluate Condition by STEM interactions, and detect smaller intervention effects.

7. References

Hojat, M. et al. Jefferson Scale of Empathy.

Spreng, R. N., McKinnon, M. C., Mar, R. A., and Levine, B. (2009). The Toronto Empathy Questionnaire: Scale development and initial validation of a factor-analytic solution to multiple empathy measures. *Journal of Personality Assessment*, 91(1), 62-71.

Higham, N. J. (1998), *Handbook of Writing for the Mathematical Sciences*, 2nd edn., Society for Industrial and Applied Mathematics, Philadelphia.

University of Manchester School of Mathematics, *Statistical Report Writing*

Oregon State University Department of Statistics, *Writing a Statistical Report for STAT 411/511*

8. Appendix

A. Index

Symbol	Meaning	Notes	Section
i	Participant index	Used throughout scoring and modeling notation	§2.6.3 , §2.6.5 , §2.6.6 , §3.4.1 , §4.4 , §4.5 , §4.7
j	Survey item index	Used in item-response notation	§2.6.3
s	Survey time index	Used for Pre/Post survey time	§2.6.3
X_{ijs}	Numeric response for participant i , item j , survey time s	Item-level score after response conversion	§2.6.3
$X_{ijs} \in \{1, 2, 3, 4, 5, 6, 7\}$	Jefferson scoring range	Equation 2.1	§2.6.3
$X_{ijs} \in \{0, 1, 2, 3, 4\}$	Toronto scoring range	Equation 2.2	§2.6.3
X	Observed numeric response	Before reverse scoring	§2.6.4
$X_{rev} = L + U - X$	General reverse-scoring formula	Applying reverse-score formula, Equation 2.3	§2.6.4
J_{ia}	Corrected Jefferson item a for participant i	After reverse scoring when needed	§2.6.5
T_{ib}	Corrected Jefferson item b for participant i	After reverse scoring when needed	§2.6.5
$J_{Total} = \sum_{a=1}^{14} J_{ia}$	Jefferson total-score construction	Equation 2.6	§2.6.5

$T_{Total} = \sum_{b=1}^{16} J_{ib}$	Toronto total-score construction	Equation 2.7	§2.6.5
$J_{Change} = J_{Post,i} - J_{Pre,i}$	Jefferson change-score construction	Equation 2.8	§2.6.6
$T_{Change} = T_{Post,i} - T_{Pre,i}$	Toronto change-score construction	Equation 2.9	§2.6.6
Y_{0ik}	Baseline/pre measurement for participant i , scale k	Used in pairing and ceiling-effect notation	§3.4.1 , §3.4.6
Y_{1ik}	Follow-up/post measurement for participant i , scale k	Used in pairing notation	§3.4.1
$\Delta_{ik} = \eta_{0k} + \eta_{1k}Y_{0ik} + \epsilon_{ik}$	Ceiling-effect model	Equation 3.3	§3.4.6
$\bar{Y}_A$ or $\bar{Y}_B$	Sample means for groups A and B	Used in Welch test	§4.1
s_A^2 or s_B^2	Sample variances for groups A and B	Used in Welch standard error	§4.1
n_A , n_B , or n	Sample sizes for groups A and B or paired observations	Used in Welch standard error and paired t-test	§4.1
s_{Δ}	Standard deviation of change scores	Used in paired t-test	§4.2
$t_W = \frac{\bar{Y}_A - \bar{Y}_B}{\sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}}}$	Welch two-sample t statistic	Equation 4.1	§4.1
Δ or Δ_i	Mean change / participant-level change score	Used in paired tests and change-score models	§4.2 , §4.4 , §4.5

$t_p = \frac{\bar{\Delta}}{s_{\Delta} / \sqrt{n}}$	Paired t-test statistic	Equation 4.2	§4.2
$\Delta_i = \beta_0 + \beta_1 G_i + \varepsilon_i$	Simple linear change-score model	Equation 4.3	§4.4
$\Delta_i = \beta_0 + \beta_1 C_i + \beta_2 S_i + \beta_3(C_i S_i) + \varepsilon_i$	Factorial change-score model	Equation 4.4	§4.5 , §5.5
$Y_{i1} = \beta_0 + \beta_1 Y_{i0} + \beta_2 C_i + \beta_3 S_i + \beta_4(C_i S_i) + \varepsilon_i$	ANCOVA model	Equation 4.6	§4.7 , §5.6
Y_{i1}	Post-intervention score	ANCOVA outcome	§4.7
Y_{i0}	Pre-intervention score	ANCOVA baseline covariate	§4.7
R^2	Model-explained variability	Reported for change-score and ANCOVA models	§5.5 , §5.6
<i>Cohen's d</i>	Standardized paired effect size	Reported for overall Pre/Post change	§5.2
<i>Cook's distance</i>	Influence diagnostic	Used to evaluate outliers	§3.4.5

B. Data Dictionary

Variable	Definition	Notes
DID #	Anonymous participant identification number	Used to match each participant's Pre and Post survey responses
Survey	Time point of survey completion	Coded as Pre or Post
Condition	Study group assignment	Indicates whether the participant was in the Intervention or Control group

STEM	STEM classification status	Indicates whether the participant was classified as STEM or Non-STEM based on major/program
J_total	Jefferson Scale total score	Total empathy score from the Jefferson-style items for a given survey response
T_total	Modified Toronto Empathy Questionnaire total score	Total empathy score from the modified TEQ items for a given survey response
J_change	Change in Jefferson total score	Calculated as $J_{\text{post}} - J_{\text{pre}}$
T_change	Change in Toronto total score	Calculated as $T_{\text{post}} - T_{\text{pre}}$
Major	Participant's reported academic major	Used to help classify participants as STEM or Non-STEM
J_pre	Participant's Jefferson total score at baseline	Pre-survey Jefferson total score
J_post	Participant's Jefferson total score after the study period	Post-survey Jefferson total score
T_pre	Participant's Toronto total score at baseline	Pre-survey modified TEQ total score
T_post	Participant's Toronto total score after the study period	Post-survey modified TEQ total score
Gender	Participant's reported gender	Demographic variable
Transfer Student	Transfer-student status	Indicates whether the participant reported being a transfer student
Upper v Lower	Academic division classification	Indicates whether the participant was classified as lower-division or upper-division
Race/Ethnicity	Participant's reported race or ethnicity	Self reported demographic variable
Class Level	Participant's academic year class level	Freshman, Sophomore, Junior, Senior, or Graduate
First Gen Status	First-generation college student status	Indicates whether the participant reported being a first-generation college student (Y/N)

C. Alluvial Plots

The project includes an *Alluvial.html* file that demonstrates how alluvial plots can be used to visualize changes in participant responses from the Pre survey to the Post survey. These plots are especially useful for Likert-scale survey data because they show how individual response categories shift over time, rather than only summarizing changes with means or test statistics. In this report, alluvial plots provide a visual way to examine whether participants moved toward higher, lower, or unchanged response categories after the science communication program.

Each alluvial plot tracks the flow of participant responses for selected survey items. The left side of the plot represents the Pre survey response category, while the right side represents the Post survey response category. The width of each flow indicates how many participants followed that response pattern.

These plots are used as exploratory visualizations rather than formal hypothesis tests. They help identify response patterns that may not be obvious from total scores alone, such as whether observed score changes are driven by many small shifts across participants or by a smaller number of large shifts.

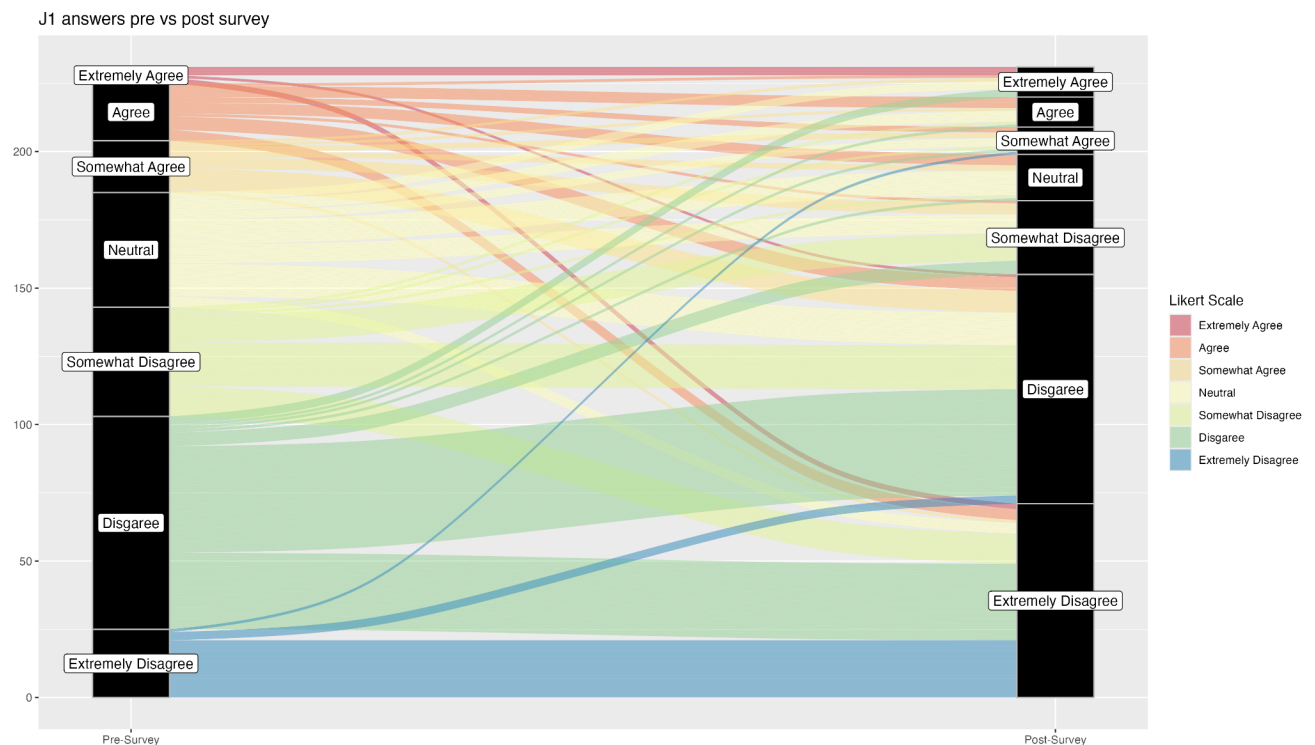


Figure 11. Alluvial plot for question “I am capable of succeeding in a career in medicine.”

D. Tutorial Notebooks

Several tutorial notebooks were developed to document and support the statistical workflow used in this report. These notebooks provide reproducible examples of data preprocessing, exploratory analysis, assumption checking, statistical testing, and model selection. They also serve as a guide for understanding how the cleaned datasets and final analysis outputs were produced.

The *Data_Preprocessing_in_R.ipynb* notebook introduces the process of cleaning and preparing the survey data in R using the tidyverse framework. This notebook covers the main data construction steps, including importing the raw survey file, removing irrelevant columns, converting text-based survey responses into numeric values, applying reverse scoring, creating total scores, standardizing major labels, assigning STEM classification, and constructing matched Pre/Post records. These steps are necessary because the raw survey responses must be transformed into consistent numerical variables before statistical analysis can be performed.

The *EDA_& Assumption_Checking.ipynb* notebook contains the exploratory data analysis and diagnostic procedures used to evaluate the structure and suitability of the data for statistical modeling. This notebook includes numerical summaries, histograms, boxplots, Q-Q plots, and checks related to independence, normality, and variance assumptions. These exploratory tools help identify distributional patterns, possible ceiling effects, outliers, and violations of modeling assumptions before formal inference is conducted.

The *Tests_Modeling_ANOVA_ANVOCA.ipynb* notebook contains examples of the main statistical tests and models used in the analysis. These include Welch's two-sample t-tests, Wilcoxon tests, paired tests, change-score comparisons, one-way and two-way ANOVA, mixed ANOVA, and ANCOVA. The notebook also provides guidance on when each test is appropriate, helping connect the research questions to the statistical methods used in the report. Together, the tutorial notebooks make the analysis workflow more transparent, reproducible, and easier to extend in future work.